

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (ISE) (Credit System) Degree Examinations
 January 2023

20ISE12 – TOTAL QUALITY MANAGEMENT FOR SUSTAINABLE GROWTH

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Define Total Quality Management and explain with a neat diagram seven phases in development of TQM.	6	L*2	1	1,2
	b) Explain Juran's quality trilogy and Juran's Formula.	6	L2	2	1,2
	c) Explain 5S HOUSE KEEPING and relation between various 5s.	8	L2	3	1,2
2.	a) Describe a framework for the improvement of a process or system using PDSA cycle and its application.	8	L2	4	1,2
	b) Discuss the EFQM Model for Excellence in small groups.	8	L2	3	1,2
	c) Discuss the three categories of the Deming Prize small groups.	4	L2	2	1,2
3.	a) What are Crosby's four absolutes of quality management and express Crosby's 14 Steps.	12	L2	3	1,2
	b) Why TQM is important to project management? List variety of product requirements and demands.	8	L2	2	1,2
Unit – II					
4.	a) Define Benchmarking and explain the concept of benchmarking.	8	L2	2	1,2
	b) What are the various approaches to benchmarking?	8	L2	3	1,2
	c) Explain the four stages of FMEA.	4	L2	2	1,2
5.	a) What difficulties are typically encountered when benchmarking direct competitors? Describe several ways to work around these problems.	12	L2	3	1,2
	b) With a neat flow diagram explain quality Function Development Process.	8	L2	4	1,2
6.	a) Explain in detail the six sigma concepts of process capability. Also bring out the methodologies adopted in six sigma practices.	7	L2	4	1,2
	b) Enlighten Six major loss areas need to be measured and tracked and give three metrics to quantify these losses.	8	L2	3	1,2
	c) What is Taguchi's Quadratic Loss Function for Nominal – the Best?	5	L2	4	1,2
Unit – III					
7.	a) What are the various measures of central tendency in quality with an example?	8	L2	4	1,2
	b) What are the benefits of ISO registration?	4	L2	5	1,2
	c) Explain the monitoring and measurement in quality system.	8	L2	5	1,2
8.	a) What are Millennium development goals of India?	8	L2	5	1,2
	b) Determine the average, median, mode, range and standard deviation for the height of seven people. Data are 1.83,1.91,1.78,1.80,1.83,1.85,1.87 meters.	4	L2	4	1,2
	c) Discuss three categories of variations in piece part production and explain control charts with example.	8	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome